



AI-Based Exam Evaluation System

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Introduction / Problem Statement



Challenges in Manual Exam Paper Evaluation

- Manual evaluation is time-consuming and labor-intensive.
- High potential for human error in grading, impacting fairness.
- Lack of consistency in marking across different evaluators.



Need for Automation

- Automation improves accuracy and reduces grading time significantly.
- Efficient handling of large volumes of answer scripts.
- Enables immediate feedback for students, enhancing learning.

Project Overview

1 System Functionality

- Upload student answers seamlessly via an easy-to-use interface.
- Generate model answers for accurate evaluation.
- Evaluate scripts using advanced LLM (LLaMA 3) to ensure reliability.

2 Technical Features

- Utilizes OCR technology for digital text extraction.
- Employs similarity matching to assess answers against model responses.
- Generates personalized feedback for students based on their performance.
- Displays marks and result status for faculty review.

Technology Components



Frontend

React.js for a responsive user interface.



Backend

Node.js and MongoDB for robust server-side processing.



AI Model

LLaMA 3 accessed via Together API or local Ollama setup.



Tools

PDF text extractor for document processing and NLP comparison logic to enhance evaluation precision.

Step-by-Step Work Flow

1

Faculty login

2

Exam Setup

3

Upload Scripts

4

AI Evaluation

5

Marks Display

6

Account History

Outcomes & Future Scope



Achievements

- Successfully implemented auto evaluation and feedback generation.
- Enabled tracking of exam history for better insights.



Future Enhancements

- Plan to integrate handwritten OCR for broader usability.
- Aim to improve accuracy and support multiple languages in evaluations.
- Enhance LLM prompts for better understanding and feedback quality.

Thank You